

Uzair Arif

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Economics & Mathematics graduate at IBA Karachi seeking Junior AI/ML & Agent Research roles. Hands-on experience designing multi-agent LLM simulations, self-evolving memory architectures, production MLOps pipelines, and time-series forecasting systems. Research intern at NCAI, FBR, and NASTP.

Education

Institute of Business Administration — BS Economics and Mathematics

June 2026

Key Courses: Foundations of Data Science, Linear Algebra, Probability Theory, Optimization Techniques, Applied Econometrics, Statistical Machine Learning, Stochastic Processes, Game Theory

Skills

Languages & ML: Python, SQL, PyTorch, Scikit-learn, XGBoost, CatBoost, SciPy, NumPy

LLMs & Agents: Multi-Agent Systems, Memory Architectures, Ollama, LangChain, RAG (FAISS/BM25/RRF)

MLOps & Tooling: Docker, FastAPI, MLflow, Pytest, Git, SQLite, Optuna, Streamlit

Core Concepts: Multi-Agent Simulation, Memory Collapse Dynamics, Time-Series Forecasting, Concept Drift

Professional Experience

National Center of Artificial Intelligence (NCAI) — Research Intern

Jul 2025 – Aug 2025

- Conducted literature review on Multi-Agent RL, public goods games, and Ostrom's governance principles; drafted a full research proposal on emergent institutional design in cooperative agent systems.
- Designed core simulation components including democratic voting on punishment rules, weighted voting schemes, and IPPO agent architecture with reward functions for norm emergence.

Federal Board of Revenue (FBR) — Revenue Forecasting Intern

Jun 2025 – Jul 2025

- Built 5 daily Prophet models (Income Tax, Sales Tax, Federal Excise, Customs Duty, Gross Collection) with Pakistan-specific Gregorian and lunar holiday effects and monthly seasonality.
- Tuned all models via Optuna hyperparameter search and evaluated on held-out validation sets using MAE and MSE to handle highly volatile government revenue patterns.
- Automated a multi-horizon Excel reporting pipeline with daily, weekly, and monthly breakdowns; deployed an interactive Streamlit dashboard to surface mid-month surges and fiscal deadline spikes for the team.

National Aerospace Science & Technology Park (NASTP) — AI Intern

Jun 2024 – Aug 2024

- Engineered real-time data streaming and visualization tools using ZeroMQ and Python, reducing publish-subscribe latency to milliseconds.
- Built CNNs, linear regression, and SVM models from scratch, improving predictive accuracy by 10%.
- Created an anomaly detection model for ECG time-series data, achieving a 95% detection rate for anomalies.

Projects

SEAM — Shared Evolving Agent Memory

- Architected a modular multi-agent simulation framework evaluating memory collapse (echo traps) and contamination across 162 factorial sweep runs using local Ollama LLMs.
- Implemented 3 memory update policies and 3 network topologies (Off, Ring, Broadcast), proving structured incremental curation achieves 55× higher task score over naive overwrite.
- Developed quantitative metric suite (Self-BLEU, embedding similarity, action entropy) tracking brevity bias and demonstrating Ring Topology mitigates organic memory poison propagation.
- Shipped 21 pytest test modules, append-only JSONL logging, experiment rehydration pipelines, and automated paper-ready Seaborn figure generation.

ELICIT: Emergent Multi-Agent Climate Institutions

- Architected a multi-agent simulation framework evaluating 26 LLM agents across 30 rounds with dual-replication protocols, simulating climate shocks, Loss-and-Damage pooling, and democratic voting.
- Ran diff-in-diff econometric models ($p < 0.05$) across 114 qualitative post-event LLM rationales ($\kappa = 0.72$), proving bad-reputation gossip causes strategic withdrawal when peer-sanctions are absent.
- Designed a risk-sharing Loss-and-Damage fund module achieving 76.8% shock coverage, coupled with an interactive HTML/JS browser visualizer for real-time agent decision and wealth telemetry.

NYC Taxi Demand Forecasting — Production MLOps System

- Built an Optuna-tuned XGBoost pipeline achieving 0.949 Val R^2 , tracked via MLflow Model Registry.
- Served predictions through FastAPI backed by a WAL-mode SQLite feature store with TTL pruning.
- Deployed KS-test drift monitoring, validated end-to-end via simulated streaming replay against the live API.
- Shipped with 36/36 tests passing, zero ruff/mypy errors, fully containerised in Docker.

IBA RAG Chatbot

- Built a modular, production-structured hybrid RAG system combining FAISS semantic search and BM25 keyword retrieval fused via Reciprocal Rank Fusion across 1000+ HTML files and PDF documents.
- Engineered custom ingestion pipeline with recursive chunking to preserve context boundaries.
- Integrated Groq LPU inference backend to achieve sub-second output generation latency across institutional documents.